

NANYANG JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

CLASS

BIOLOGY

9648/02

Paper 2 Core Paper

September 2015

Additional Materials: Answer Paper

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
8	
Section B	
Total	100

This document consists of **21** printed pages and **1** blank page.

[Turn over

Section A

Answer **all** the questions in this section.

- 1 **Fig. 1.1** is a computer-generated image of the enzyme hexokinase binding with its substrate, glucose. The product of the enzyme-catalysed reaction is glucose-6-phosphate.

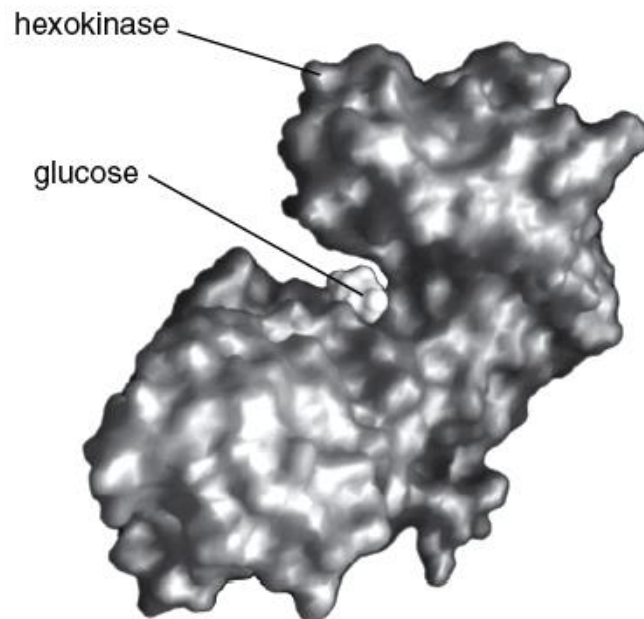


Fig. 1.1

- (a) Hexokinase binds with glucose using the induced fit mechanism.
Describe how an enzyme-substrate complex forms by this mechanism.

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[3]

- (b) Suggest how enzymes which use the induced fit mechanism can be less affected by competitive inhibitors than those which use the lock and key mechanism.

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[2]

- (c) Glucose can enter cells by active transport or facilitated diffusion.
Glucose-6-phosphate is a molecule that cannot move out of cells.

(i) Describe two **differences** between active transport and facilitated diffusion.

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[2]

(ii) Suggest why glucose-6-phosphate cannot move out of cells.

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[2]

[Total: 9]

- 2 **Fig. 2.1** is a section through the rough endoplasmic reticulum representing the synthesis of proteins.

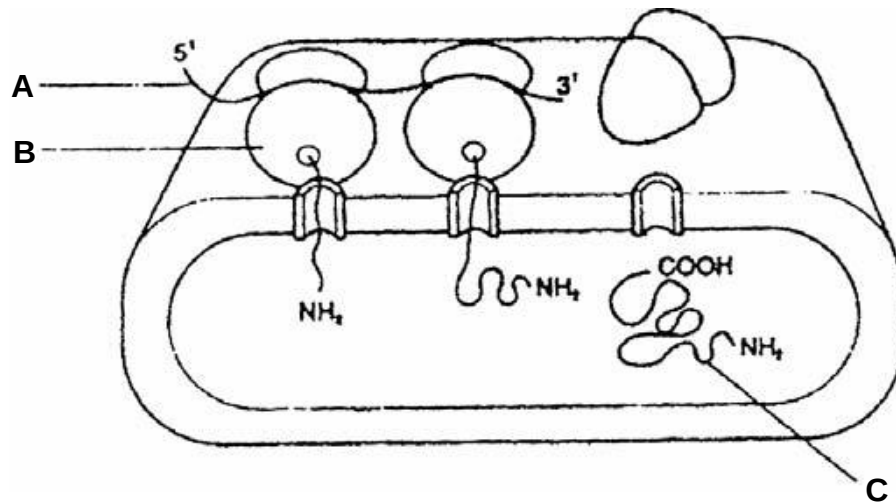


Fig. 2.1

- (a) Name the structures **A** to **C**.

A

B

C

[3]

- (b) Describe and explain the role of structure **B** in protein synthesis.

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[3]

Fig. 2.2 is a diagram of a section of mRNA showing the sequence of three of the codons.

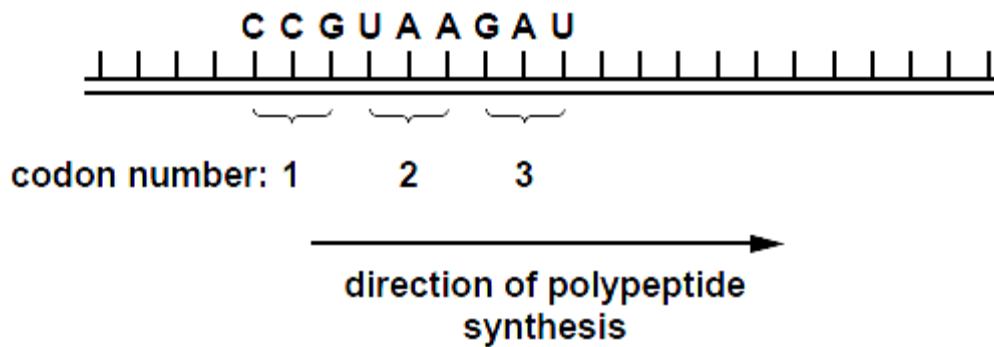


Fig. 2.2

(c) State the base sequences of

(i) the tRNA anticodon complementary to codon 1.

..... [1]

(ii) the DNA sequence which coded for codon 3.

..... [1]

(d) Explain the relationship between the sequence of the DNA triplets and the structure of the hormone.

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 [2]

[Total: 10]

- 3 Viruses such as bacteriophages have been described as being “organisms at the edge of life”.

(a) State one characteristic of viruses that may classify them as being

(i) living.

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[1]

(ii) non-living.

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[1]

Lambda (λ) phage infections result in the lysis of the bacterial host cells which shows up as a plaque or a zone of clearing on an agar plate containing a dense growth of bacteria. This observation can be used to quantify the number of infectious bacteriophage particles through a plaque assay. 'Plaque Forming Unit (PFU)' gives the measurement of the number of infectious viral particles present during the experiment.

Fig. 3.1 below shows the changes in PFU caused by lambda (λ) phages infecting E.coli cells upon irradiation by UV.

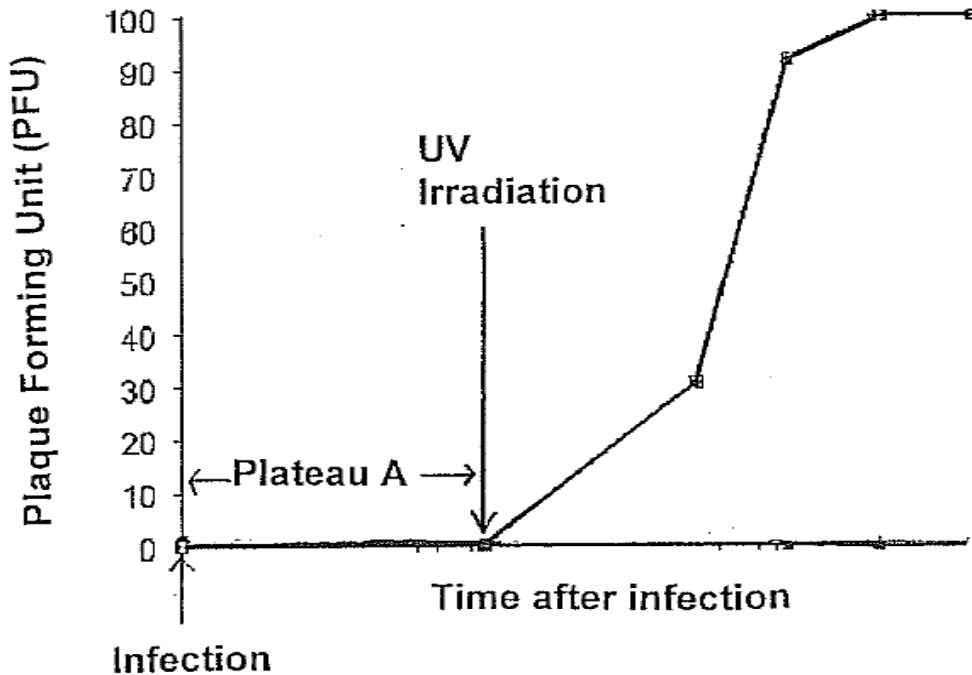


Fig. 3.1

(b) Briefly describe the structure of a lambda (λ) phage.

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[3]

(c) Suggest an explanation for the flat part of the graph, labelled A, as shown in **Fig. 3.1**.

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[2]

(d) Account for the rise in PFU shortly after UV irradiation.

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[3]

[Total: 10]

- 4 The Ras protein, encoded by the Ras proto-oncogene, is a G protein that relays a signal from a growth factor receptor on the plasma membrane to a cascade of protein kinases. The conversion of the Ras proto-oncogene to Ras oncogene causes the latter to act in a dominant manner. The permanently activated protein directs the cell to grow and divide in the absence of external signals, leading to uncontrolled cell division and the formation of a tumour.

Somatic mutations in the H-ras gene in bladder cells have been associated with bladder cancer. One specific gene mutation has been identified in a significant percentage of bladder tumors. A segment of the gene is shown in **Table 4**.

Table 4

codon number	9	10	11	12	13	14	15
normal Human H-Ras	GUG Val	GGC Gly	GCC Ala	GGC Gly	GGU Gly	GUG Val	GGC Gly
oncogenic H-Ras	GUG Val	GGC Gly	GCC Ala	GUC Val	GGU Gly	GUG Val	GGC Gly

- (a) With reference to **Table 4**, explain how this mutation would affect the **function** of the protein formed.

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[3]

- (b) Explain what is meant by “the Ras oncogene acts in a dominant manner”.

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[2]

- (c) Besides gene mutation, describe one way in which a proto-oncogene may be converted to oncogene.

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[2]

- (d) The expression of H-Ras gene can be furthered controlled at post-translational level. Explain how gene expression may be controlled at post-translational level.

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[3]

- (e) Many individuals are afflicted with cancers when they are past 40 years old. Suggest why these cancers do not normally appear in younger individuals.

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[2]

[Total: 12]

5

- (a) Explain what is meant by the term *heterozygous genotype*.

heterozygous:

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genotype:

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[2]

- (b) The budgerigar, *Melopsittacus undulatus*, is a small type of parrot that is native to Australia.

Fig. 5.1 shows a budgerigar.



Fig. 5.1

A budgerigar can have blue, green, yellow or white feathers.

Two genes, **A / a** and **D / d**, are involved in the inheritance of feather colour in budgerigars.

- A bird has at least one dominant allele **A** but is homozygous for **d** has blue feathers.
- A bird which has at least one dominant allele **D** but is homozygous for **a** has yellow feathers.
- A bird with at least one dominant **A** allele and one dominant **D** allele has green feathers.
- A bird that is homozygous for **a** and **d** has white feathers.

Two green-feathered budgerigars, heterozygous at both gene loci, were crossed. Draw a genetic diagram of this cross to show the probability of producing offspring with yellow feathers.

[4]

(c) State the name for this type of interaction between gene loci.

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[1]

- (d) Explain how you would determine that a budgerigar with green feathers is homozygous dominant for both gene loci.

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[2]

[Total: 9]

6 Fig. 6.1 outlines some steps in glucose metabolism in mammalian cells.

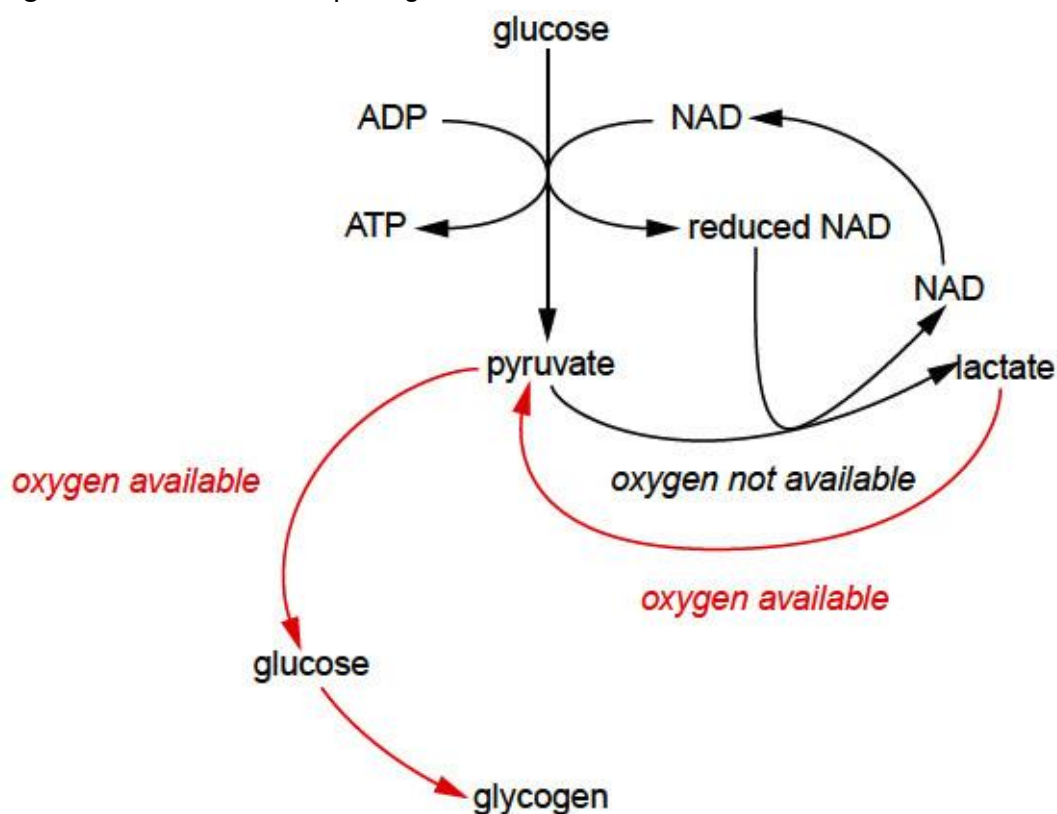


Fig. 6.1

(a) With reference to Fig. 6.1,

(i) explain why, in the absence of oxygen, pyruvate needs to be converted to lactate;

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[2]

(ii) name the type of reaction **and** the type of bonds formed when glucose molecules are used to make glycogen.

Type of reaction:

Type of bonds:

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[1]

- (b) ATP is often known as the universal 'energy currency'.
Outline how ATP is suitable for acting as an energy currency.

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[2]

- (c) Most ATP is made in cells by membrane systems that create proton gradients by pumping protons from one compartment to another.

Fig. 6.2 shows two such membrane systems.

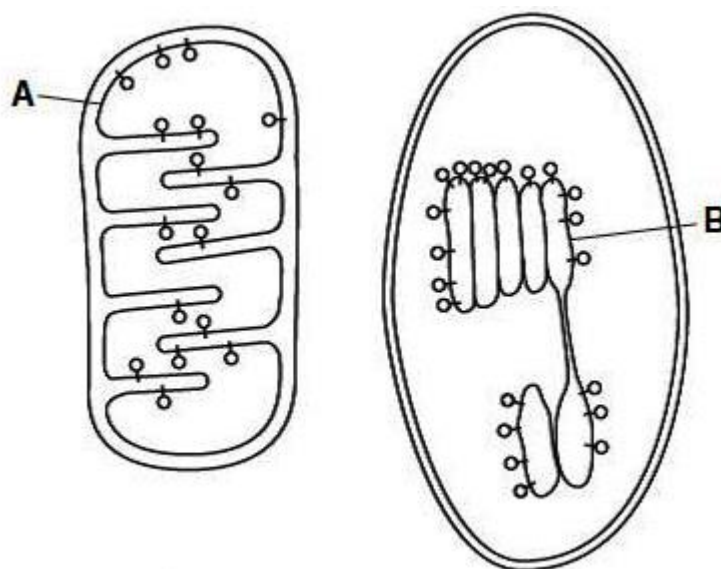


Fig. 6.2

- (i) Draw arrows onto each of the membrane systems in **Fig. 6.2** to show the direction in which protons are pumped. [1]
- (ii) Outline how energy is made available for pumping protons across such membranes.

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[3]

(iii) Explain how ATP synthase is involved in the production of ATP.

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[2]

[Total: 11]

- 7 Serotonin is a neurotransmitter which is produced by certain neurons in the brain. One of its effects is to increase the activity of sensory neurons in the brain. It also usually improves a person's mood and keeps the person awake.

Fig. 7 below shows a synapse which serotonin is the neurotransmitter.

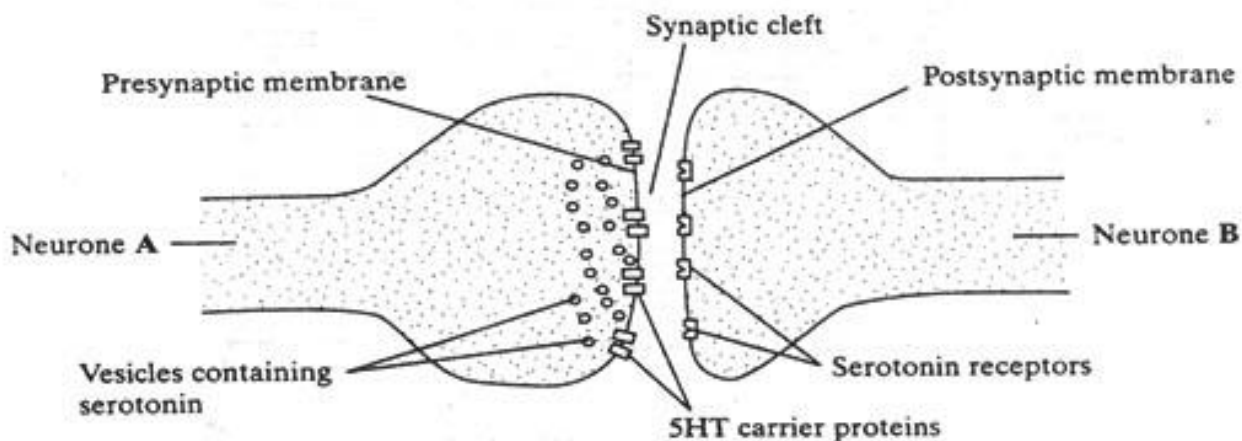


Fig. 7

(a)

- (i) Describe how the neurotransmitter is released into the synaptic cleft.

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[2]

- (ii) Explain how the release of the neurotransmitter, serotonin, results in the transmission of an impulse from neuron A to neuron B.

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[4]

Serotonin is normally rapidly reabsorbed from the synaptic cleft by 5HT carrier proteins in the presynaptic membrane.

(b) Suggest an advantage of rapidly absorbing the serotonin.

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[1]

The active ingredient in the drug, Ecstasy, is MDMA. It blocks the attachment of serotonin molecules to the 5HT carrier proteins.

(c) Suggest how MDMA may temporarily improve a person's mood

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[2]

[Total: 9]

- 8 There are over 40 Galapagos Islands including the small and isolated island named Daphne Major. Only two species of Darwin finches are found on this island. From 1975 to 1977, Daphne Major experienced a severe drought. A ground finch, *Geospiza fortis*, feeds on seeds on Daphne Major.

Fig. 8.1 shows the ground finch, *G. fortis*.



Fig. 8.1

- One of the few plants that survived the drought produced large seeds inside tough fruits.
 - Many *G. fortis* died during the drought and the population declined from 1400 in 1975 to 190 in 1977.
 - The mean beak depth of those *G. fortis* that died was 10.68 mm and the mean beak depth of those that survived was 11.07 mm.
 - The mean beak depth of *G. fortis* before the drought was 10.86 mm.
- (a) Calculate the percentage decrease in population size between the years 1975 and 1977. Show all the steps in your calculation and give your answer to the nearest whole number.

[2]

- (b) Describe the environmental factors that act as forces of natural selection on finches during drought years.

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[2]

(c) Suggest why some *G. fortis* were able to survive the drought while others died.

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[3]

Natural selection was taking place on Daphne Major.

(d) Suggest why the type of natural selection operating on *G. fortis* **during** the drought is directional selection.

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[1]

(e) Outline the mechanisms that may have let natural selection lead to the evolution of the thirteen species of Darwin finches now found on the Galapagos Islands.

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[2]

[Total: 10]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 9 (a)** Outline the arrangements and functions of named membranes within the cell. [7]
- (b)** Explain how the fluid mosaic model of cell surface membrane facilitates the transport of substances. [7]
- (c)** Describe the main stages of cell signaling using a named example. [6]

[Total:20]

- 10 (a)** Describe how the prokaryote, E.coli, is able to respond to varying concentrations of lactose and glucose [8]
- (b)** Explain how genetic variation in bacteria could have arisen. [8]
- (c)** Suggest why the influenza virus displays high genetic diversity. [4]

[Total:20]